

The spill-mat design lab

Teacher guide and worked answers

Plan a fair material comparison, record water-retention results and justify a product choice.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Spring | B31:C31

Plan a fair test to compare material properties; record results.

Directly develops the stated science outcome. Original secondary-age exemplar activities; diagnostic-led support and stretch follow the SoW. No claim of qualification approval.

Prior learning

Materials have different properties and must be selected for a purpose.

Change the tested sample while keeping the method consistent.

Subtract two masses; use a calculator or provided scaffold if useful.

Success evidence

Name the changed factor, measured outcome and at least two controls.

Record wet and dry mass, then calculate retained water with units.

Choose a product using the stated requirement and acknowledge a test limitation.

Equipment

Complete paper route: shared design brief, constructed evidence table, chosen response route, pencil and optional calculator.

Optional practical: prepared one-layer 10 cm × 10 cm samples of kitchen paper, cotton cloth and felt; three fresh pieces per product.

Optional practical: digital balance reading 0.1 g, dry weighing tray, room-temperature tap water in a stable shallow tray, timer and blunt tongs.

Preparation

Print the shared page. The brief and product costs are fictional teaching values, not market prices.

If using the optional practical, prepare sample sizes, test that each can be handled wet, and practise drying/resetting and taring the weighing tray before every measurement. Weigh wet samples promptly after the fixed drip time.

Use enough water to immerse each sample fully. Replace or top up consistently; no sample should be partly dry because the tray runs low.

Set out separate wet and dry areas and provide paper towels for spills. If the scale resolution or available time is unsuitable, use the complete supplied evidence route.

Practical care

Use only small amounts of room-temperature tap water and clean prepared samples. No tasting, detergents, heating or unknown liquids.

Keep the battery-powered scale on a stable dry area. Carry wet samples over the water tray; wipe spills promptly.

Use blunt tongs and prepared squares; teacher handles cutting. Avoid splashing and follow the school's normal eye-protection arrangements if required.

Never place the test mat by electrical equipment or use it as real spill-safety equipment. The product brief is a classroom model.

Ready to teach alternative

The default route audits a test plan and analyses supplied constructed measurements. Optional groups collect a small actual comparison only with pre-set apparatus; record actual results separately.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Design a removable spill-mat insert
2	Retrieval	2-5	Connect properties to a purpose
3	I do	5-13	What are we actually testing?
4	I do	Within stage	Subtract the starting mass
5	I do	Within stage	Keep the question and method matched
6	We do	13-21	Rescue this unfair test plan
7	We do	Within stage	Read a repeat, not a promise
8	We do	Within stage	Best for which job?
9	Hinge	21-24	What makes the comparison fair?
10	You do	24-34	Write the product team's report
11	You do	Within stage	Record the result you have
12	You do	Within stage	Keep the conclusion within the evidence
13	Review	34-38	Challenge the design decision
14	Review	Within stage	Plan our next improvement
15	Exit	38-40	Exit: the fair-test check

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Design a removable spill-mat insert

Introduce a fictional club drinks-station mat insert tested on a tray away from any equipment. It is a material-selection model, not a certified safety product. Read the every-trial requirement deliberately; an average alone does not meet this brief.

2 Retrieval Connect properties to a purpose

R1 takes liquid into its structure; accept soaks up then refine. R2 soak time stays the same, product type changes. R3 not necessarily: subtract dry mass. Diagnose language versus numerical difficulty separately and provide a calculator if needed.

3 I do What are we actually testing?

8 minutes shared across three I do slides. Point to spaces between fibres. The diagram is a simplified structure model: water enters connected spaces and is retained through interactions at small scales. The blue marks are not oversized single molecules. We measure a practical water-retention outcome, which includes method-dependent remaining water.

4 I do Subtract the starting mass

Shared I do time. Think aloud: I cannot count the cloth itself as water. I measured its dry mass first, so I subtract that from the wet mass. The answer is a mass of retained water in grams. Use a calculator openly; arithmetic is a tool for answering the science question.

5 I do Keep the question and method matched

Shared I do time. Explain that our question compares particular one-layer 10 cm squares of three products. Different thickness and construction are part of these products, so we cannot isolate the intrinsic material alone. Control the water conditions, immersion and handling as well. Before every measurement, dry/reset and tare the weighing tray. Weigh dry, immerse, drip vertically without squeezing, then weigh wet promptly.

6 We do Rescue this unfair test plan

8 minutes shared across three We do slides. Pairs act as design reviewers. Expected fixes: equal 10 cm square area, one layer, same 30 s full immersion, 10 s vertical drip, no squeezing, dry mass first and wet mass after. Fresh dry samples for each repeat; same scale and water conditions. All pupils identify at least two specific repairs.

7 We do Read a repeat, not a promise

Shared We do time. Paper does not meet threshold; cotton and felt do. Note that repeated values are not identical, which is normal variation. The values support this small test, not a promise of all future performance.

8 We do Best for which job?

Shared We do time. Cotton is lowest cost among passing products. Felt retains more water in these trials but costs more. Untested properties include wet strength, reusability, drying time, slip and durability; we have not evaluated those. Distinguish design criterion from most absorbent.

Reveal: Cotton meets all three trials and costs less than felt.

9 Hinge What makes the comparison fair?

Collect choices independently. B is correct. For A or D revisit the method controls; for C explain that fresh dry samples keep the starting condition comparable. All routes then choose a plan, record results and decide.

A. Use different sizes because products differ. - No. More area can change water retained; the product-size question would be different.

B. Use equal-area dry samples and one fixed method. - Yes. Keep relevant conditions consistent while changing the product.

C. Reuse each wet sample for every repeat. - No. This changes the starting wetness. Use fresh dry pieces for these repeats.

D. Squeeze only the cotton before weighing. - No. Different handling changes how much water remains, so it confounds the comparison.

10 You do Write the product team's report

10 minutes shared across three You do slides. All pupils use same question and objective; support varies recording and numerical demands. Optional actual test uses pre-set apparatus: a pair may test one product with fresh dry repeats while other pairs test other products. Collate only if conditions matched. Default paper dataset provides all evidence within time.

11 You do Record the result you have

Shared You do time. For a torn or dropped sample record the event rather than silently discarding. A retest can be justified but must be identified. Pupil activity pages use constructed data; never claim the class obtained those values. Provide a worked subtraction on a separate example if arithmetic blocks the science.

12 You do Keep the conclusion within the evidence

Shared You do time. Expected cotton decision with 6.8 g minimum and 4 credits, compared with felt at 6 credits. Supported pupils may use any correct qualifying trial plus awareness all passed. Stretch route should distinguish means from every-trial thresholds and rethink testing per gram versus per square.

13 Review Challenge the design decision

4 minutes shared over two Review slides. Both observations may be true; cotton answers this brief. Ask pupils to state a different brief for which felt might be favoured, such as maximum retained mass per same-area square with no cost limit. Avoid declaring a universal winner.

14 Review Plan our next improvement

Light Lundy loop: invite private written, spoken or pointed suggestions; record and explain the next actionable test. Example repeated-use durability measured after equal soak/dry cycles. Do not promise a particular timetable. Pupil voice improves the investigation design.

15 Exit Exit: the fair-test check

E1 product type; retained-water mass after method. E2 any relevant controls such as area and soak time. E3 cotton: every result at least 6 g, cheapest passing product at 4 credits; numerical example minimum 6.8 g. Collect one independent plan/evidence sentence per pupil.

Answers and assessment

R1-R3

R1 absorbent means takes liquid into its structure. R2 keep soak time the same; change product type. R3 no: the wet mass includes the dry product, so subtract dry mass to find retained water.

W1

Repair equal sample area at 10 cm × 10 cm, one layer, fixed 30 s full immersion, fixed 10 s vertical drip without squeezing, dry mass before and wet mass after, same water conditions and scale. Use fresh dry samples for repeats. Two specific relevant fixes establish initial fair-test understanding.

W2 / We do interpretation

Paper retained 5.2, 5.5, 5.3 g: fails 6 g requirement. Cotton retained 7.0, 6.8, 7.2 g: passes every observed trial. Felt retained 8.0, 7.8, 8.2 g: passes. Cotton is cheapest passing product at 4 credits, versus felt 6 credits. Felt holds more, but maximum absorption is not this brief's only criterion.

Hinge A-D

A incorrect because area changes the comparison. B correct: equal-area dry samples with a fixed method compare these products. C incorrect because wetness at the start changes; use fresh dry samples. D incorrect because squeezing changes retention and introduces inconsistent treatment.

S1-S4

S1 product type. S2 retained water mass in g. S3 any two: area, soak time, drip time, one layer, water conditions, scale/handling; not the final result. S4 paper $6.0 - 0.8 = 5.2$ g; cotton $9.0 - 2.0 = 7.0$ g; felt $10.5 - 2.5 = 8.0$ g.

S5

Cotton; 6; felt. Stronger explanation mentions its lowest observed result is 6.8 g, all three results meet the threshold, and 4 credits is less than 6 credits. Do not choose paper simply because it is cheapest: it fails the performance requirement.

N1-N2

N1 change product; measure water retained after fixed soak/drip method; control area, layer count, immersion time, drip time, handling, water conditions and scale. Three valid controls sufficient. N2 measure fresh dry mass, fully immerse 30 s, drip vertically 10 s without squeezing, measure wet mass, subtract dry mass, repeat using three fresh dry pieces per product.

N3

Paper T1/T2/T3: 5.2/5.5/5.3 g. Cotton: 7.0/6.8/7.2 g. Felt: 8.0/7.8/8.2 g. Example cotton trial 2: $8.8 - 2.0 = 6.8$ g. Units and correct subtraction matter more than unaided arithmetic.

N4-N5

N4 cotton: minimum 6.8 g exceeds 6 g in these trials, costs 4 credits versus felt 6. N5 these are three trials of particular products, equal area but differing thickness/structure; other mats or criteria may differ. Next test could compare more repeats, repeated-use integrity, drying time or wet strength using matched procedures. No claims about slip or safety have been tested.

T1

Retained masses as N3. Means: paper $(5.2 + 5.5 + 5.3)/3 = 5.333\dots$ g, about 5.3 g; cotton $21.0/3 = 7.0$ g; felt $24.0/3 = 8.0$ g. Accept sensible consistent rounding.

T2

The extra set averages $(5 + 7 + 9)/3 = 7$ g, but the 5 g trial fails the 6 g requirement. Thus an average above 6 g cannot establish every observed trial meets 6 g. Even three passing trials do not guarantee all future trials pass.

T3-T4

T3 cotton is lowest-cost passing product. Uncertainty: small sample, product variation, handling, balance precision. Improve with more matched fresh samples and fixed timing/handling; report variation. T4 the test compares equal-area, one-layer pieces of those products for this insert. Retained water per gram describes mass efficiency: using means, paper about $5.33/0.8 \approx 6.7$ g/g; cotton $7/2 = 3.5$ g/g; felt $8/2.5 = 3.2$ g/g. It is a different question and may rank products differently; it does not replace the stated same-area threshold brief.

T5

Accept a safe measurable second property. Example reusability: after equal soak/drip/dry cycles, record whether each same-size sample remains intact using a defined criterion; control cycle count and handling,

use repeats. Or drying time: after the same saturation/drip protocol weigh at fixed intervals under the same air conditions until a stated mass criterion is reached. Teacher reviews feasibility before any follow-on practical.

E1-E4 / actual record

E1 product type; mass of water retained in g. E2 any two relevant controls with reason, e.g. equal area because larger pieces may hold more. E3 cotton, all trials at least 6 g, minimum 6.8 g, cheapest passing cost 4 credits. E4 a specific improvement linked to evidence; record tears/drops and justified retests openly. Actual results have no fixed answer: assess accurate recording and valid interpretation. Reflection accepts any evidence-linked choice.

Assessment / re-teach

Secure objective: identifies changed factor and measured outcome; specifies at least two meaningful controls; subtracts dry mass correctly with support if needed; records units; makes a brief-matched evidence decision. If choosing heaviest wet sample, revisit the dry-versus-wet worked example. If choosing felt only because greatest retention, reread lowest cost that meets every trial. If claiming all felt is best, ask which products and sizes were actually tested.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 8, We do 8, hinge 3, independent 10, review/Lundy 4, exit 2. Zero-minute slides share stage time.

This is an original secondary-age product-design investigation aligned to Spr1-W4. It applies material-property thinking from the two conductivity lessons while introducing absorbency as a distinct testable property.

The mass-retention protocol measures water remaining after fixed conditions, not a universal intrinsic absorption constant. Surface water and structure can contribute; matched handling matters.

We intentionally compare equal-area one-layer products, with their actual thickness and construction. Do not imply thickness was controlled or that the result isolates material composition. A per-mass or equal-thickness investigation is a different follow-on question.

Default evidence route is complete. For an optional practical, preassemble equipment and restrict the sample count to fit the 10-minute task; pupils still make an independent design conclusion and complete the exit.

Do not claim school-collected observations from the constructed table. A failed, torn or inconsistent actual sample is evidence to discuss, not a result to hide.

Access and responsive teaching

Supported

Use page 2 with explicit change/measure/keep choices and a subtraction frame. Read numerical values aloud; use calculator, pointing or dictated explanations.

Standard

Use page 3 to specify a method, calculate three first-trial retained masses, interpret repeat evidence and justify the cheapest product that meets the requirement.

Stretch

Use the Stretch route to calculate means, compare the every-trial requirement with an average, and redesign the test for a different question. Same fair-test and evidence objective.

Regulation

Offer dry investigator, timer, recorder or paper analyst roles. No forced water handling. Keep the sequence visible and allow a short planned pause. Assess science separately from handwriting and group confidence.

Misconceptions to check

The heaviest wet sample absorbed the most water.

Subtract each sample's dry mass. Different products can start with different masses.

Check: Compare the dry and wet values in the worked example.

A fair test means every material must give the same result.

A fair comparison keeps relevant conditions the same; results may differ because of the tested sample.

Check: Ask which is controlled: soak time or final water mass.

Repeating a wet sample is the same as testing a fresh dry one.

A wet starting sample changes the starting condition. Use fresh dry samples for these repeats.

Check: Ask why squeezing a used sample does not restore the original condition.

The top result proves all products made from that material are best.

This comparison is of particular one-layer products at a stated size. Thickness and construction differ and the brief may value other properties.

Check: Ask whether a thicker different cotton product could behave differently.

Word	Meaning
absorbent	Able to take in a liquid into its structure.
retain	Hold after a stated waiting or dripping time.
fair comparison	A comparison with relevant conditions kept the same so the result can answer the question.
variable	A factor that can change in an investigation.
repeat	Carry out the same test again to check variation.

Scientific references

[American Chemical Society: Designing an Absorbency Test](#)

Checked 9 September 2026. Supports comparing absorbency with consistent conditions and a simple structure explanation. Our original mass-retention method, secondary-age product brief, diagrams and data are newly authored.

[Department for Education: Science programmes of study](#)

Checked 9 September 2026. Upper-KS2 materials content supports evidence-based reasons for uses of materials and comparative/fair tests, matching the stated GROW SoW spine. All lesson tasks are original.